

(%i1) **a:matrix([1,2],[3,4]);**

a
$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

(%i2) **echelon(a);**

(%o2)
$$\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

(%i3) **rank(a);**

(%o3) 2

(%i6) **b:matrix([4,5],[6,7]);**

b
$$\begin{bmatrix} 4 & 5 \\ 6 & 7 \end{bmatrix}$$

(%i7) **echelon(b);**

(%o7)
$$\begin{bmatrix} 1 & \frac{5}{4} \\ 0 & 1 \end{bmatrix}$$

(%i8) **rank(b);**

(%o8) 2

(%i9) **c:matrix([1,2,3],[7,8,9],[3,4,5]);**

c
$$\begin{bmatrix} 1 & 2 & 3 \\ 7 & 8 & 9 \\ 3 & 4 & 5 \end{bmatrix}$$

(%i10) **echelon(c);**

(%o10)
$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

(%i11) **rank(c);**

(%o11) 2

(%i12) **d:matrix([4,7,8],[9,10,11],[12,13,14]);**

d
$$\begin{bmatrix} 4 & 7 & 8 \\ 9 & 10 & 11 \\ 12 & 13 & 14 \end{bmatrix}$$

(%i13) **echelon(d);**

(%o13)
$$\begin{bmatrix} 1 & \frac{7}{4} & 2 \\ 0 & 1 & \frac{28}{23} \\ 0 & 0 & 1 \end{bmatrix}$$

(%i14) **rank(d);**

(%o14) 3

(%i15) **e:matrix([1,2,3,4],[5,6,7,8],[9,10,11,12],[7,7,3,3]);**

e

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 7 & 7 & 3 & 3 \end{bmatrix}$$

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(%i16) echelon(e);
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(%o16)

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

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(%i17) rank(e);
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(%o17) 3
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